

Retention Radar: The Drivers of Customer Churn and What to Do Next

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1) Executive Summary

This analysis examined customer churn patterns and developed a predictive risk scoring model to proactively identify customers at risk of attrition. Using demographic, service, contract, and billing variables, a logistic regression model was trained to estimate individual churn probabilities. The model achieved a **ROC–AUC of 0.84**, indicating strong ability to distinguish between churners and retained customers.

At a glance (model + targeting)

Model performance: ROC–AUC = 0.84
Top 10% risk segment: ~75% precision, ~28% recall
Top 30% risk segment: ~57% precision, ~65% recall
Highest-risk segment: month-to-month + electronic check (53.7% churn)

Risk-based segmentation shows that churn can be efficiently targeted. Focusing on the **top 10% highest-risk customers** yields approximately **75% precision**, meaning three out of four targeted customers are true churn risks, while capturing approximately 28% of total churners. Expanding outreach to the top 20–30% risk segment significantly increases churn capture (up to 65%) while maintaining acceptable precision levels. This provides operational flexibility depending on retention budget and intervention capacity.

Key structural drivers of churn include:

- Month-to-month contract type
- Short tenure (0–3 months)
- Manual or electronic cheque payment methods
- Fiber customers without value-added service bundles
- Higher monthly charges without perceived added value

Based on these insights, a targeted retention strategy is recommended. Priority initiatives include migrating month-to-month customers to annual contracts, incentivizing automatic payment enrollment, strengthening onboarding for early-tenure customers, bundling support/security services for fiber users, and deploying risk-based outreach to the highest predicted churn segment.

Overall, the predictive model enables a transition from reactive churn response to proactive, data-driven retention management. By aligning intervention resources with high-risk segments, the organization can reduce attrition, improve customer lifetime value, and maximize retention return on investment.

2. Context and Use Cases

2.1 Why customer churn matters

Customer churn matters because it directly affects recurring revenue, profitability, and growth efficiency. When a customer churns, the business loses future monthly cash flows and often needs to spend more on marketing and sales to replace that lost revenue. High churn also reduces customer lifetime value (CLV), which makes acquisition campaigns less profitable and can force the company to raise prices or cut costs to maintain margins.

Operationally, churn creates instability. It increases workload for customer service and retention teams, drives more cancellation-related inquiries, and makes demand forecasting harder. If churn risk is identified early, teams can focus retention offers and support on the customers most likely to leave, rather than offering discounts broadly and eroding margin.

Strategically, churn measurement protects the customer base and guides product decisions. Understanding which factors (such as month-to-month contracts, certain payment methods, or specific service bundles) increase churn risk helps prevent concentration in fragile segments, supports better packaging and pricing, and improves long-term retention through targeted improvements (e.g., onboarding, service reliability, or support quality).

2.2 How to use this analysis operationally

- **Retention targeting** — identify high-risk customers and prioritize outreach, service recovery, or incentives.
- **Contract strategy** — design and test offers to move customers from month-to-month to longer-term contracts.
- **Billing optimization** — reduce churn linked to payment friction by promoting auto-pay or alternative payment methods.
- **Product bundling** — evaluate whether add-ons (OnlineSecurity, TechSupport, Backup) reduce churn and where bundling works best.
- **Onboarding and early-life programs** — reduce churn in the first 1–3 months by improving activation, education, and support.
- **Service quality monitoring** — focus investigation on high-churn product groups (e.g., fiber customers) to identify experience gaps.

3. Problem Statement and Objectives

Build and evaluate a baseline churn risk model using tenure, contract type, services subscribed (internet and add-ons), and billing/payment attributes to predict customer churn, then translate results into actionable recommendations to reduce churn and protect recurring revenue.

3.1 Objectives

- **Quantify overall churn prevalence** and assess class imbalance (Churn: Yes vs No)

- **Identify how churn varies** by key segments:
 - tenure bands (e.g., 0–3, 4–12, 13–24, 25–48, 49+ months)
 - contract type (Month-to-month vs One/Two year)
 - payment method (Electronic check vs auto-pay methods vs mailed check)
 - internet service type (DSL vs Fiber optic vs None)
- **Compare churners vs non-churners** on key commercial drivers (MonthlyCharges, TotalCharges) and engagement proxies (PaperlessBilling)
- **Assess product stickiness** by testing whether add-ons (OnlineSecurity, TechSupport, Backup, DeviceProtection, Streaming) are associated with lower churn, overall and within internet type
- **Train a baseline model** and evaluate performance using AUC, precision, recall, and threshold trade-offs to support practical targeting
- **Translate results into actions** (retention plays, packaging changes, billing interventions) prioritized by expected impact and feasibility

Data Description

This dataset contains **7,044 customer records** with **21 variables**, designed to analyze **customer churn** in a subscription telecom context. Each row represents a unique customer account, identified by customerID. The target variable is Churn, indicating whether the customer **left the service** (Yes) or **remained active** (No) at the point the record was captured.

The dataset contains **customer demographics**, **account tenure**, **service subscriptions**, and **billing/payment attributes**. These features help describe both customer profile and product/billing experience, which are common drivers of churn behavior.

Dataset size and quality checks

- **Rows:** 7,044
- **Columns:** 21
- **Duplicates:** None detected (based on customerID)
- **Missing values:** None detected in the exported file
- **Timeline:** No explicit timestamp field is provided, so analysis is **cross-sectional** rather than time-series. tenure is used as the main time-related variable (months with the company).

Column Groups and Definitions

1) Customer identifiers and demographics

- **customerID:** Unique identifier for each customer account (non-predictive; used for indexing).
- **gender:** Customer gender (Male/Female).
- **SeniorCitizen:** Indicator variable (0 = not senior citizen, 1 = senior citizen).
- **Partner:** Whether the customer has a partner (Yes/No).
- **Dependents:** Whether the customer has dependents (Yes/No).

2) Relationship duration (time with the company)

- **tenure:** Number of months the customer has stayed with the company. This is typically one of the strongest churn predictors; churn often concentrates in early tenure periods.

3) Core services

- **PhoneService:** Whether the customer has phone service (Yes/No).
- **MultipleLines:** Whether the customer has multiple phone lines (Yes/No/No phone service).
- **InternetService:** Type of internet service (DSL/Fiber optic/No).

4) Optional add-on services (often linked to “stickiness”)

- **OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport:** Add-on protection/support services (Yes/No/No internet service).
- **StreamingTV, StreamingMovies:** Entertainment add-ons (Yes/No/No internet service).

5) Contract and billing configuration

- **Contract:** Contract term (Month-to-month/One year/Two year). This is a key business lever: longer contracts usually reduce churn.
- **PaperlessBilling:** Whether the customer uses paperless billing (Yes/No).
- **PaymentMethod:** Payment channel (Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic)).

6) Charges and revenue proxies

- **MonthlyCharges:** Current monthly billed amount for services (continuous).
- **TotalCharges:** Total amount charged over the customer’s lifetime (continuous; typically increases with tenure, so it is strongly correlated with tenure).

7) Target variable

- **Churn:** Whether the customer churned (Yes/No). This is the primary outcome variable for segmentation and predictive modeling.

Analytical Considerations and Assumptions

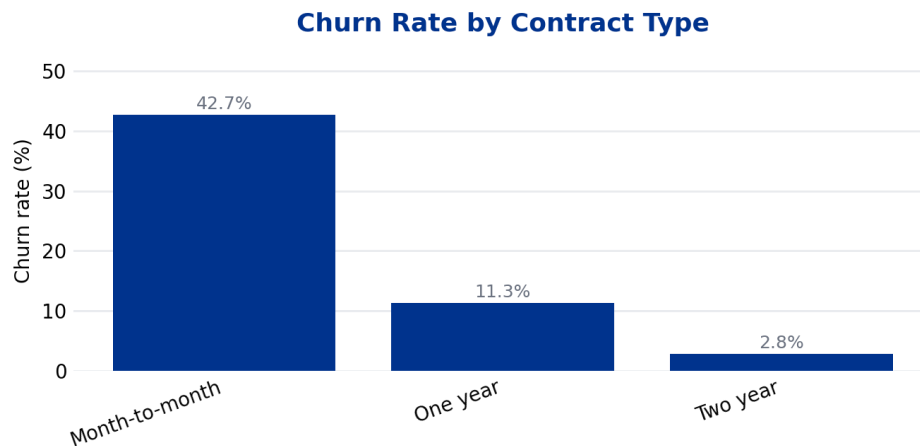
- Because the dataset has **no event date** for churn, churn is treated as a **binary outcome at snapshot time**, not a survival/time-to-event measure.
- TotalCharges is partially determined by tenure and MonthlyCharges, so it may introduce redundancy. For modeling, it should be tested for multicollinearity and potential proxy effects.

- Several service fields include “**No internet service**” or “**No phone service**” as values; these should be treated as meaningful categories rather than missing data.

4) Segment Analysis

A) Contract & billing

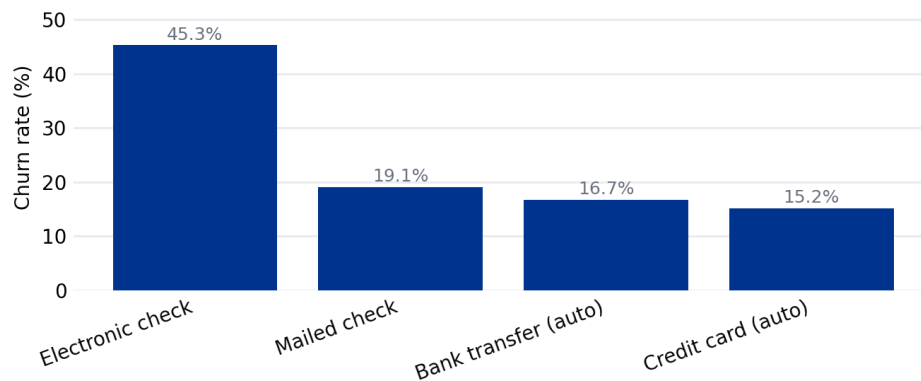
- **Contract**



Churn varies sharply by contract type. Customers on **month-to-month contracts** have a churn rate of **42.7%**, compared to **11.3%** for **one-year** contracts and **2.8%** for **two-year** contracts. This indicates contract length is a key retention lever: month-to-month customers churn at nearly **4×** the rate of one-year customers and roughly **15×** the rate of two-year customers. Operationally, churn reduction efforts should prioritize migrating month-to-month customers to longer-term contracts through targeted incentives and improved retention journeys.

- **Payment Method**

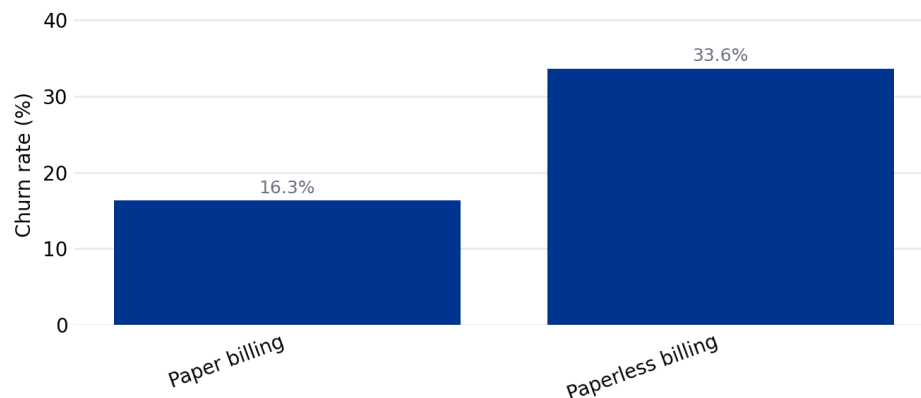
Churn Rate by Payment Method



Churn differs substantially by payment method. Customers paying via **electronic check** churn at **45.3%**, significantly higher than customers using **automatic payments** (15.2% for credit card autopay and 16.7% for bank transfer autopay) or **mailed check** (19.1%). Electronic check customers churn at roughly **3×** the rate of autopay customers, making payment method a key operational lever. This pattern suggests churn may be influenced by billing friction, payment reliability, or the type of customer segment associated with electronic check usage.

- **Paperless Billing**

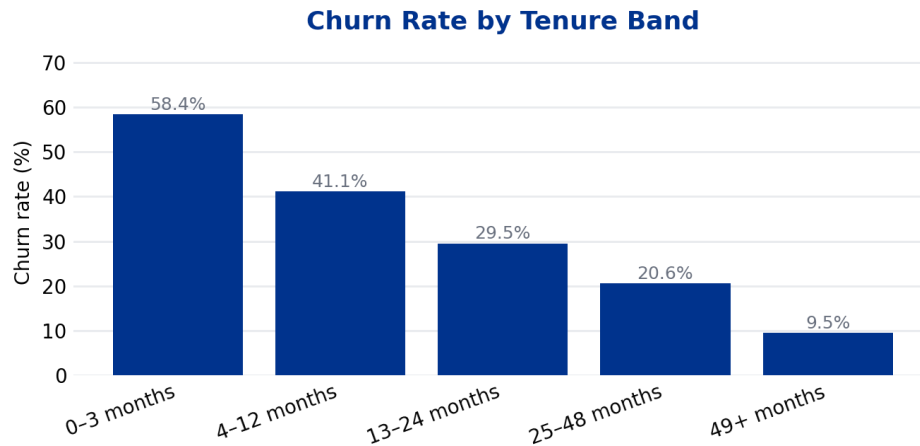
Churn Rate by Paperless Billing



Churn differs strongly by billing format. Customers enrolled in **paperless billing** churn at **33.6%**, compared to **16.3%** among customers who are not paperless. This is a **17.2 percentage point** difference (approximately **2× higher churn**) for paperless customers. Since paperless billing often correlates with digital payment channels and month-to-month contracts, this segment should be analyzed together with **payment method** and **contract type** to identify the highest-risk billing configurations.

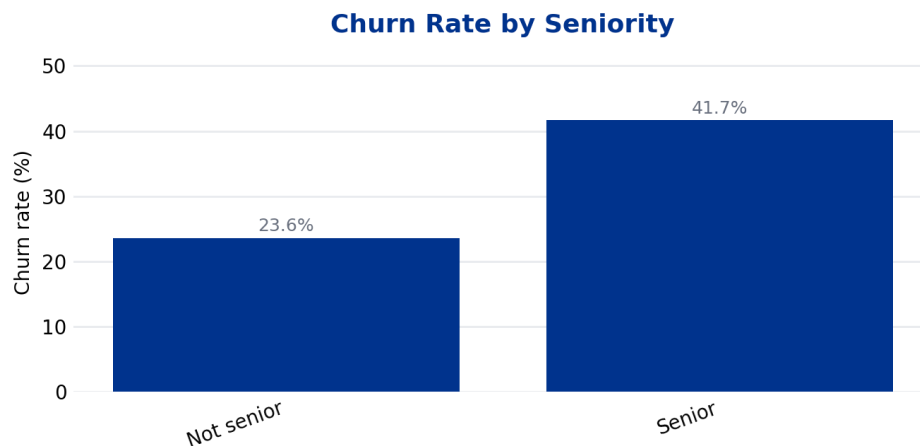
B) Tenure & customer profile

- **Tenure bands**



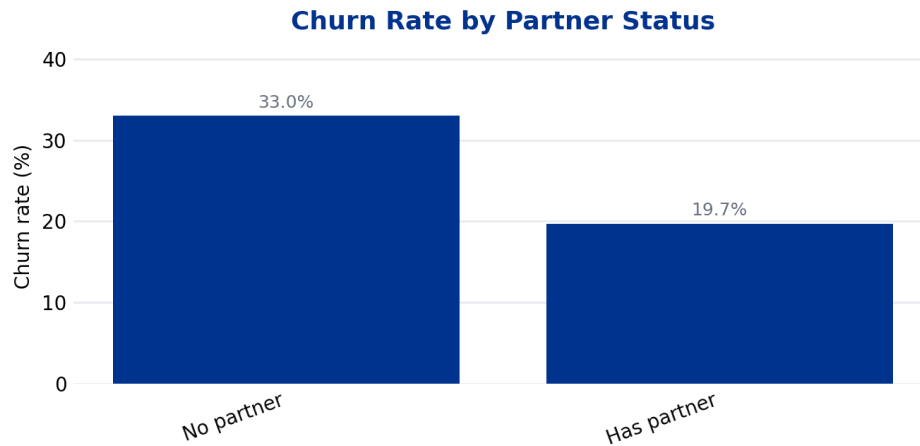
The overall churn in the dataset is **27%**. Churn is strongly concentrated among newer customers: customers with tenure ≤ 3 months have a churn rate of **58.4%**, which declines consistently across tenure bands to **41.1%** (4–12 months), **29.5%** (13–24 months), **20.6%** (25–48 months), and **9.5%** for customers with **49+ months** of tenure. This indicates that churn is primarily an **early-lifecycle problem**, and retention efforts should prioritize customers in their first year and especially the first three months where churn risk is highest.

- **Senior Citizen**



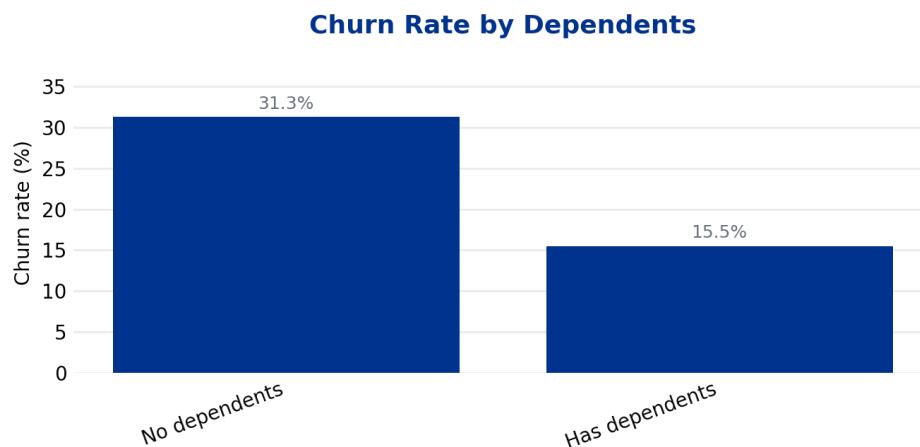
Churn varies substantially by seniority status. Non-senior customers have a churn rate of **23.6%**, while senior customers churn at **41.7%**. This indicates seniors are a higher-risk segment, with churn roughly **18 percentage points higher** (about **1.8×** the churn rate) compared to non-seniors

- **Partner**



Churn differs materially by partnership status. Customers **without a partner** churn at **33.0%**, compared to **19.7%** for customers **with a partner**. This is a **13.3 percentage point** increase (approximately **1.7× higher churn**) for customers who do not have a partner, suggesting that household stability may be associated with stronger retention

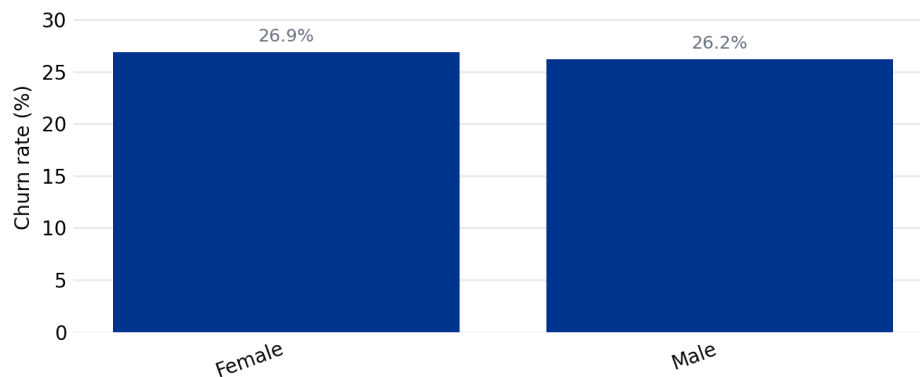
- **Dependents**



Churn varies significantly by dependent status. Customers **without dependents** have a churn rate of **31.3%**, while customers **with dependents** churn at **15.5%**. This represents a **15.8 percentage point** difference, meaning customers without dependents are roughly **2× more likely to churn**. This suggests customers with dependents may have more stable household needs and therefore stronger retention

- **Gender**

Churn Rate by Gender

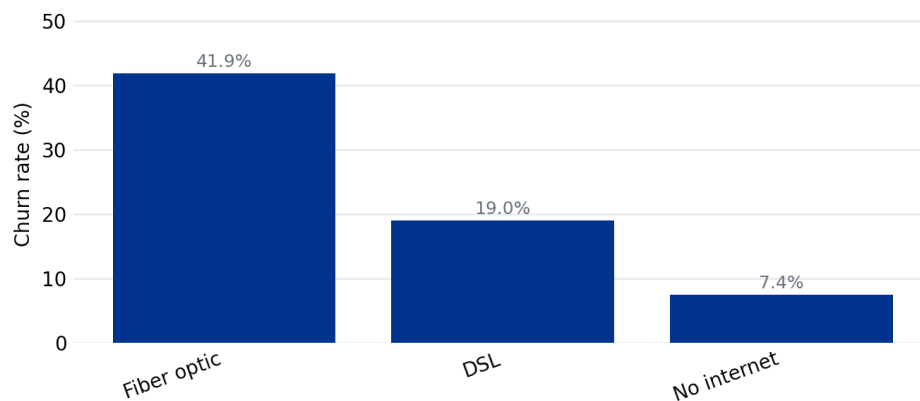


Churn rates are nearly identical across gender. Female customers churn at **26.9%** (939 of 3,488), while male customers churn at **26.2%** (930 of 3,555). The difference is only **0.7 percentage points**, suggesting gender is not a meaningful driver of churn in this dataset.

C) Product / service configuration

- **InternetService**

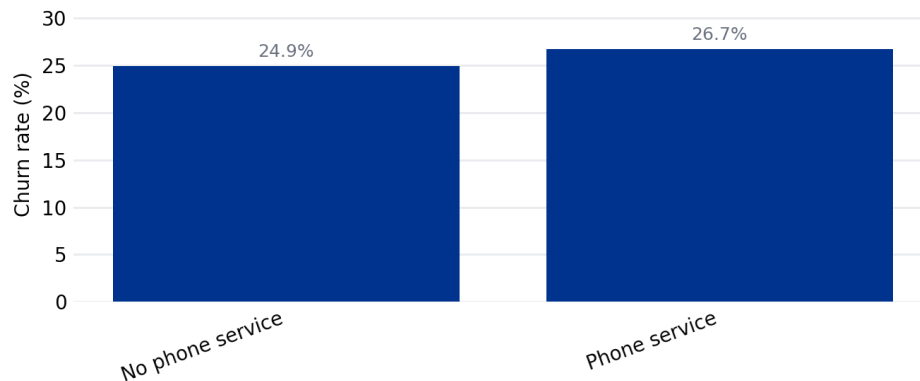
Churn Rate by Internet Service Type



Churn varies significantly by internet service type. Customers on **fiber optic** internet churn at **41.9%**, compared to **19.0%** for **DSL** and **7.4%** for customers with **no internet service**. Fiber customers churn at over 2× the rate of DSL customers, indicating that fiber service is associated with substantially higher churn risk. This may reflect differences in pricing (higher MonthlyCharges), customer expectations, or service experience and should be investigated further through contract and payment method interactions.

- **PhoneService**

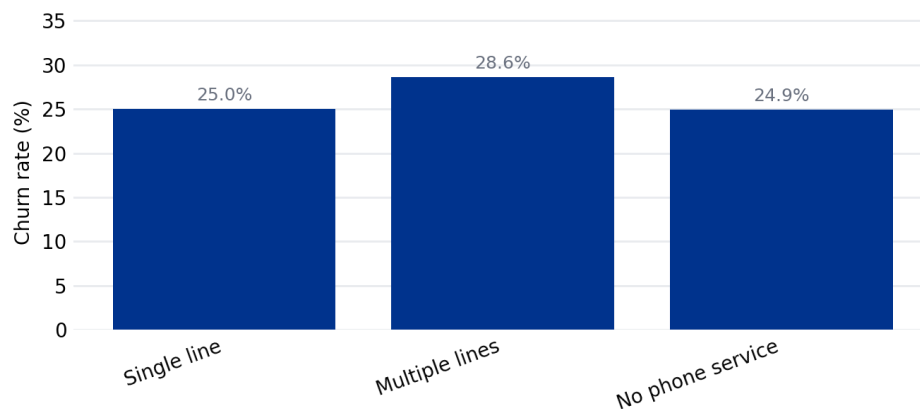
Churn Rate by Phone Service



Churn rates are very similar for customers with and without phone service. Customers without phone service churn at **24.9%**, compared to **26.7%** for customers with phone service—a difference of **1.8 percentage points**. This suggests phone service alone is not a primary churn driver compared to factors like contract type, payment method, tenure, and internet service type

- **MultipleLines**

Churn Rate by Multiple Lines



Churn shows a modest variation by multiple line status. Customers with **multiple lines** churn at **28.6%**, compared to **25.0%** for customers with a single line and **24.9%** for customers with no phone service. The difference is approximately **3.7 percentage points**, indicating that multiple lines is a secondary churn correlate and likely interacts with stronger drivers such as contract type, payment method, and internet service.

Contract × Payment Method × Churn

Contract type and payment method interact strongly in explaining churn. Month-to-month customers represent **55.0%** of the customer base and exhibit substantially higher churn across all payment methods. The highest-risk segment is **month-to-month customers paying via electronic check**, with a churn rate of **53.7%**. This segment is also large (1,850 customers, approximately **26% of the total**

base) and therefore contributes disproportionately to overall churn. Across all payment methods, churn decreases sharply for longer contracts (one-year and two-year), while electronic check remains the highest-churn payment method within each contract tier. Operationally, this identifies a clear priority segment for retention: month-to-month electronic check customers

Contract x Internet Service Provider x Churn

Contract type and internet service interact strongly in explaining churn. The highest-risk segment is **month-to-month fiber optic customers**, with a churn rate of **54.6%**. Churn decreases substantially for longer contracts across all internet types; however, **fiber optic customers consistently churn at higher rates** than DSL and no-internet customers within each contract tier (e.g., **19.3% vs 9.3%** under one-year contracts). This indicates that fiber churn risk is driven both by contract structure and by factors associated with the fiber service experience (such as pricing, expectations, or service quality).

Tenure band x Contract x Churn

Tenure and contract type interact strongly in explaining churn. For **month-to-month customers**, churn is highest in early tenure and declines with time, but remains elevated even among long-tenure customers (e.g., **60.1%** churn for tenure band 1 and **26.0%** for tenure band 5). In contrast, customers on **one-year contracts** show substantially lower churn across tenure bands (approximately **6.9%–13.3%**), and **two-year contracts** exhibit the lowest churn overall (near-zero churn in early tenure bands and **2.1%–3.3%** in later tenure). This confirms that churn is primarily driven by a combination of **early lifecycle risk** and **contract structure**, with month-to-month customers representing the highest-priority retention group.

8) Predictive Risk Scoring

Predictive Churn Risk Model

To proactively identify customers at risk of attrition, a supervised machine learning model was developed using logistic regression. The model was trained to predict the probability of churn using demographic attributes, service characteristics, contract details, billing behavior, and usage-related features. The target variable was binary (Churn: Yes/No), and the dataset was split into training and test subsets using stratified sampling to preserve class balance. Data preprocessing included imputation of missing values, numeric scaling, and one-hot encoding of categorical variables to ensure model stability and interpretability.

Model performance was evaluated using the Area Under the Receiver Operating Characteristic Curve (ROC–AUC), which measures the model’s ability to distinguish between churners and non-churners. The model achieved a **ROC–AUC of 0.8414**, indicating strong discriminatory power and reliable ranking capability. This level of performance suggests the model is well-suited for risk scoring and customer prioritization.

Risk-Based Targeting Strategy

Rather than using a fixed probability threshold, customers were ranked by predicted churn probability and segmented into risk percentiles. This enables flexible targeting strategies depending on operational capacity and intervention cost.

When targeting the **top 10% highest-risk customers**, the model achieved:

- **Precision: 75.2%** (approximately three out of four targeted customers are true churners)
- **Recall: 28.3%** (captures roughly 28% of total churners)

Expanding the target group improves churn capture at the expense of precision:

- **Top 20% targeting** captures approximately **49.7% of churners** with **66.0% precision**
- **Top 30% targeting** captures approximately **65.0% of churners** with **57.4% precision**

These results demonstrate a clear precision–recall trade-off. A narrower targeting strategy maximizes efficiency (higher hit rate), while broader targeting increases churn prevention coverage.

Business Implications

The model enables proactive churn mitigation by identifying high-risk customers before attrition occurs. Customers in the highest risk decile exhibit predicted churn probabilities exceeding 90%, making them strong candidates for targeted retention interventions such as contract conversion incentives, service upgrades, loyalty offers, or proactive support outreach.

Operationally, the appropriate targeting threshold should be determined by balancing intervention cost against expected retention value. If retention actions are resource-intensive, focusing on the top 10% risk segment ensures high efficiency. If interventions are automated or low-cost, expanding outreach to the top 20–30% may significantly reduce overall churn volume.

Overall, the predictive risk model provides a scalable, data-driven framework for churn prevention and strategic customer retention planning.

9) Recommendations & Action Plan

Strategic Retention Recommendations

Based on the segmentation analysis and predictive risk scoring model (AUC = 0.84), the following targeted retention strategies are recommended. Each initiative is designed to address identified churn drivers and high-risk customer segments.

1. Contract Migration Strategy

Target Segment (Who):

Month-to-month customers, particularly those in the top 20% churn risk segment and tenure below 12 months.

Intervention (What):

Introduce incentives to migrate customers from month-to-month contracts to 12- or 24-month plans. Incentives may include limited-time discounts, loyalty perks, free service add-ons (e.g., StreamingTV), or price lock guarantees.

Rationale (Why it works):

Contract type is a strong predictor of churn. Month-to-month customers face minimal switching barriers and exhibit significantly higher churn rates. Longer-term contracts increase switching costs and commitment, stabilizing retention.

KPI to Monitor (How to measure):

- Conversion rate from month-to-month to annual contracts
- 90-day post-conversion churn rate
- Churn rate differential between contract types
- Incremental revenue retained from converted customers

2. Automatic Payment Adoption Program

Target Segment (Who):

Customers using electronic check or non-automatic payment methods, especially those in high-risk deciles.

Intervention (What):

Offer small bill credits or loyalty points for enrolling in automatic payments (credit card or bank auto-pay). Simplify enrollment during customer service interactions and digital account management.

Rationale (Why it works):

Manual payment methods are often associated with billing friction and higher churn. Automatic payments reduce late payments, improve customer stickiness, and create habitual engagement.

KPI to Monitor:

- Auto-pay enrollment rate
- Churn rate by payment method
- Billing-related service calls
- Payment failure rate

3. Fiber Customer Service Bundle

Target Segment (Who):

Fiber internet customers without TechSupport or OnlineSecurity add-ons, particularly those in the top 30% risk segment.

Intervention (What):

Bundle TechSupport and OnlineSecurity services at a discounted rate for fiber customers. Offer a 3-month free trial of support/security services to high-risk customers.

Rationale (Why it works):

Fiber customers often have higher monthly charges and may experience service sensitivity. Value-added services increase perceived utility, reduce dissatisfaction, and improve service dependency.

KPI to Monitor:

- **Bundle adoption rate**
- Churn rate among fiber customers with vs. without add-ons
- Customer satisfaction scores (CSAT)
- Average revenue per user (ARPU)

4. Early-Life Onboarding Program

Target Segment (Who):

Customers with tenure between 0–3 months, particularly in high-risk deciles.

Intervention (What):

Launch a structured onboarding journey including welcome emails, service education content, proactive check-ins, and a satisfaction survey within the first 60 days.

Rationale (Why it works):

Early-tenure customers show elevated churn risk. The first 90 days are critical for habit formation and perceived value. Proactive engagement reduces confusion, increases product understanding, and strengthens customer attachment.

KPI to Monitor:

- 90-day churn rate
- Engagement rate with onboarding content
- Early-life NPS or satisfaction scores
- First 3-month retention lift

5. Risk-Based Retention Outreach

Target Segment (Who):

Customers in the top 10–20% predicted churn risk segment.

Intervention (What):

Deploy proactive outbound retention campaigns (personalized email, SMS, or call center outreach) offering targeted incentives based on customer profile (contract upgrade, add-on bundle, or pricing adjustment).

Rationale (Why it works):

The predictive model demonstrates strong ranking ability (AUC 0.84). Targeting the top 10% risk segment yields ~75% precision, meaning most contacted customers are genuine churn risks. Focused intervention maximizes retention ROI.

KPI to Monitor:

- Offer acceptance rate
- Post-intervention churn rate
- Cost per retained customer
- Retention ROI (revenue saved vs. campaign cost)

6. High-Charge Customer Value Reinforcement

Target Segment (Who):

Customers with high MonthlyCharges in mid-tenure stages (12–36 months).

Intervention (What):

Introduce loyalty rewards, bundled discounts, or exclusive service upgrades to reinforce perceived value relative to price paid.

Rationale (Why it works):

Customers with higher monthly charges may be more price-sensitive and more likely to churn if perceived value declines. Reinforcing value reduces the likelihood of competitive switching.

KPI to Monitor:

- Churn rate among high-charge segment
- ARPU retention
- Upgrade adoption rate
- Net revenue retention

Prioritization Framework

If intervention capacity is limited, prioritize as follows:

1. Top 10% churn risk segment (high precision, high ROI)

2. Month-to-month customers
3. Early-tenure (0–3 months)
4. Fiber customers without add-ons
5. Manual payment method customers

This tiered strategy balances efficiency and churn capture volume.

Strategic Conclusion

The predictive risk scoring model enables a shift from reactive churn management to proactive retention intervention. By targeting high-risk customers and addressing structural churn drivers, that is contract flexibility, payment friction, early dissatisfaction, and perceived service value the organization can meaningfully reduce churn while optimizing retention investment efficiency.